

# PLANT MAINTENANCE & OEE SOLUTION

*End-to-End Asset & Equipment Management Platform for Automotive Manufacturing*

↓ Reduced Downtime · ↑ Improved OEE · 🏭 Industry 4.0 Ready

## KEY FEATURES

### 🔧 Asset & Equipment Mgmt

360° registry with A/B/C criticality, machine master, OEM details & tooling lifecycle.

### 🔧 Preventive Maintenance

Auto-scheduled PMs, OEM-aligned checklists, e-signature, photo capture & mobile execution.

### 🔧 Breakdown Maintenance

Instant mobile alerts, auto escalation matrix, MTTR/MTBF tracking & structured CAPA.

### 🔧 Predictive Maintenance

AI failure prediction via IoT sensors, vibration & temperature analysis, RUL estimation.

### 🔧 Work Order & Spare Parts

Digital PM/BM/CM/IM work orders with BOM linkage, smart reorder alerts & cost tracking.

### 🔧 Dashboards & Live KPIs

Real-time OEE, MTBF, MTTR, PM%, downtime Pareto & cost — mobile-ready & MES-linked.

## Key KPIs to Track

Before and After Implementation Measure:

| KPI                   | ✔️ Target Improvement |
|-----------------------|-----------------------|
| UnPlanned Downtime    | ↓ 20-50%              |
| MTTR                  | ↓ 10-30%              |
| MTBF                  | ↑ 15-40%              |
| OEE                   | ↑ 3-8%                |
| Maintenance Cost/Unit | ↓ 10-25%              |
| Spare Inventory       | ↓ 15-25%              |
| PM Compliance         | ↑ > 90%               |

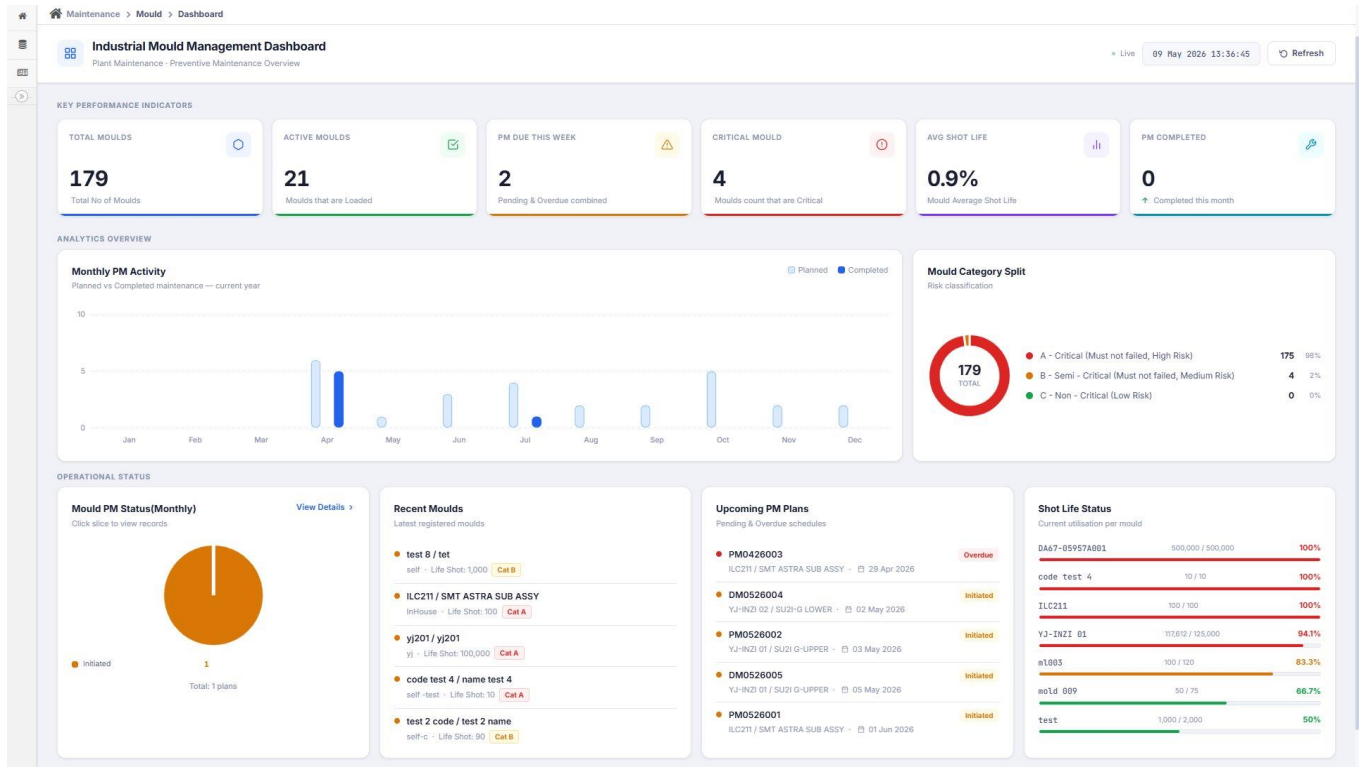
## WHY LARCH? KEY ADVANTAGES

|                                                                                                                                    |                                                                                                                                             |                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>↓ 20–40% Downtime Reduction</b></p> <p>Predictive &amp; preventive actions eliminate unplanned stops before they happen.</p> | <p><b>↑ OEE Equipment Effectiveness</b></p> <p>Real-time availability, performance &amp; quality tracking drives measurable throughput.</p> | <p><b>↑ Life Extended Machine Life</b></p> <p>Data-driven routines and early failure detection maximise asset lifespan.</p>          |
| <p><b>↓ Cost Optimised Maintenance</b></p> <p>Smarter spare planning, scheduled PM &amp; fewer emergencies lower total spend.</p>  | <p><b>✓ Audit IATF 16949 / ISO Ready</b></p> <p>Digital e-signed records and instant reports eliminate audit preparation stress.</p>        | <p><b>⚙️ 4.0 Industry 4.0 Native</b></p> <p>IoT, AI anomaly detection &amp; MES integration for the smart factory of the future.</p> |

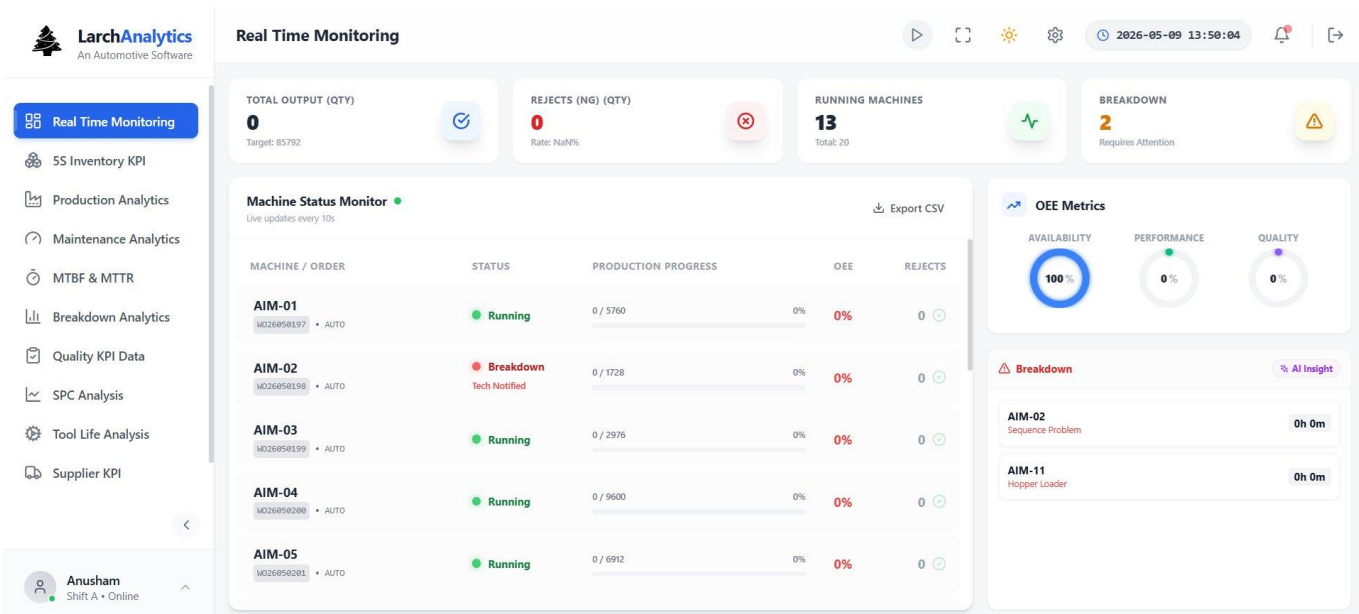
## MANUAL vs LARCH: SIDE BY SIDE

| Area              | ✗ Manual System                    | ✓ Larch Platform                          |
|-------------------|------------------------------------|-------------------------------------------|
| Work Orders       | ✗ Paper-based / job cards          | ✓ Digital, auto-assigned work orders      |
| Breakdown Alert   | ✗ Phone calls, manual entry        | ✓ Instant mobile alerts & ticketing       |
| PM Scheduling     | ✗ Calendar reminders, often missed | ✓ Automated time / usage-based PM         |
| Machine History   | ✗ Registers & paper files          | ✓ Centralised, searchable digital records |
| Spare Parts       | ✗ Excel / manual stock check       | ✓ Real-time inventory with smart alerts   |
| Downtime Analysis | ✗ Difficult & severely delayed     | ✓ Live downtime Pareto dashboards         |
| KPIs (MTTR/MTBF)  | ✗ Hard or impossible to calculate  | ✓ Auto-generated, always up to date       |
| OEE Monitoring    | ✗ Not possible easily              | ✓ Real-time OEE + MES integration         |
| Audit Readiness   | ✗ Time-consuming paperwork         | ✓ Instant digital reports, e-signature    |
| IoT / Predictive  | ✗ Completely impossible            | ✓ AI + sensor-based failure prediction    |

## LIVE DASHBOARDS — SEE LARCH IN ACTION



**Industrial Mould Management Dashboard** — Real-time KPI cards, PM compliance tracking, shot life status & criticality classification (A/B/C) across all moulds.



**Real-Time Monitoring** — Live machine status per shift, OEE metrics (Availability / Performance / Quality), active breakdowns with AI-generated failure insights.

## KEY KPIS TRACKED IN REAL TIME

|                                               |                                           |                                    |                                   |                                      |
|-----------------------------------------------|-------------------------------------------|------------------------------------|-----------------------------------|--------------------------------------|
| <b>OEE</b><br>Overall Equipment Effectiveness | <b>MTBF</b><br>Mean Time Between Failures | <b>MTTR</b><br>Mean Time To Repair | <b>PM %</b><br>PM Compliance Rate | <b>Cost</b><br>Maintenance vs Budget |
|-----------------------------------------------|-------------------------------------------|------------------------------------|-----------------------------------|--------------------------------------|

## FINANCIAL IMPACT: MANUAL vs LARCH

| Cost Factor          | ✗ Manual System                    | ✓ Larch Platform                       |
|----------------------|------------------------------------|----------------------------------------|
| Downtime Cost        | High — untracked production loss   | Reduced 20–40% via PM & predictive     |
| Emergency Repairs    | Frequent, expensive, unplanned     | Minimised via early failure alerts     |
| Spare Parts Spend    | Uncontrolled — overstock/stockouts | Optimised with smart reorder alerts    |
| Audit & Compliance   | High — manual paperwork overhead   | Lower — instant digital reports        |
| Maintenance Labour   | Low efficiency, reactive only      | High efficiency — planned & predictive |
| Machine Life         | Reduced — run-to-failure           | Improved — data-driven care routines   |
| Overall Productivity | Lower — frequent disruptions       | Higher — consistent planned uptime     |

## RETURN ON INVESTMENT

|                                                                                  |                                                                                  |                                                                                       |                                                                               |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>20–40%</b><br><b>Downtime Reduction</b><br>Fewer breakdowns & faster response | <b>↑ OEE</b><br><b>Production Uptime</b><br>More output from the same asset base | <b>↓ Spend</b><br><b>Maintenance Cost</b><br>Smarter planning reduces emergency spend | <b>100%</b><br><b>Audit Compliance</b><br>Zero prep — always inspection-ready |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|

## Ready to Achieve Zero-Breakdown Manufacturing?

*Let Larch transform your plant maintenance — smarter, faster, fully digital.*

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